

Bisphosphonates

Bisphosphonates are drugs that in certain situations can help to protect your bones against some of the effects of cancer, such as pain and weakness.

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What are bisphosphonates?

Bisphosphonates are drugs that strengthen bones and reduce the risk of bones breaking (fracture). They can also be used to treat a high level of calcium in the blood (hypercalcaemia). They may also help relieve bone pain.

You may have bisphosphonate therapy:

- if your cancer treatment causes bone thinning ([osteoporosis](#)). For example, if you are having treatment that causes [early menopause](#) or [hormone therapy for prostate cancer](#)
- to treat bone problems caused by [secondary bone cancer](#)
- to prevent or treat bone damage caused by [myeloma](#)
- to lower the risk of early [breast cancer](#) spreading to the bone.

There are different types of bisphosphonates. The 1 you have will depend on your general health and the type of cancer you have.

Types of bisphosphonates

Zoledronic acid

[Zoledronic acid](#) is a type of bisphosphate that is given by a drip (infusion). It can be used to:

- help protect the bones from the effects of some treatments for early [breast cancer](#)
- reduce the risk of early breast cancer coming back – this is called adjuvant treatment
- reduce a raised calcium level in the blood caused by cancer that has spread to the bones ([secondary bone cancer](#))
- treat bone weakness or pain caused by [myeloma](#) or by breast cancer that has spread to the bones.

Pamidronate

Pamidronate is a type of bisphosphosphate that is given by a drip (infusion). It can be used to treat:

- high levels of calcium in the blood caused by cancer that has spread to the bones
- bone weakness or pain caused by myeloma or by breast cancer that has spread to the bones.

Clodronate

Clodronate is a type of bisphosphosphate that you take as capsules or tablets. It can be used to treat:

- high levels of calcium in the blood caused by cancer that has spread to the bones
- bone weakness or pain caused by myeloma or breast cancer that has spread to the bones.

Ibandronic acid

Ibandronic acid is a type of bisphosphosphate that is given by drip (infusion) or as a tablet. It can be used to treat:

- high levels of calcium in the blood caused by cancer that has spread to the bones
- bone weakness or pain caused by breast cancer that has spread to the bones.

Bisphosphonates side effects

Some common side effects of bisphosphonates include:

- flu-like symptoms, such as pains in the muscles and joints, which usually settle after the first dose
- low levels of calcium in the blood – your doctor may advise you to take calcium and vitamin D supplements to prevent this
- shortness of breath
- diarrhoea
- indigestion or a sore throat – usually more common in people who are taking oral bisphosphonates

Your doctor can give you drugs to help control some side effects. It is important to take the drugs exactly as your specialist or pharmacist explains. This means they will be more likely to work for you. Your nurse will give you advice about managing your side effects. After your treatment is over, most side effects start to improve.

Jaw problems (osteonecrosis)

Rarely, these treatments can affect the jawbone. Healthy bone in the jaw becomes damaged and dies. This is called osteonecrosis of the jaw (ONJ). It can cause:

- pain
- loosening of the teeth
- problems with the way the gums heal.

The risk of jaw problems is higher:

- after some types of dental treatment
- in people who have gum disease
- if dentures do not fit well.

It is important to avoid having any dental treatment that could affect your jawbone when you are having bisphosphonates. This includes having a tooth or root removed, or dental implants put in. You can still have fillings, gum treatments, or a scale and polish. Your healthcare team can tell you more.

About our information

References



Below is a sample of the sources used in our bisphosphonates information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

von Moos R, Costa L, Gonzalez-Suarez E, Terpos E, Niepel D, Body JJ. Management of bone health in solid tumours: From bisphosphonates to a monoclonal antibody. *Cancer Treat Rev.* 2019 Jun;76:57-67.

Porta-Sales J, Garzón-Rodríguez C, Llorens-Torromé S, Brunelli C, Pigni A, Caraceni A. Evidence on the analgesic role of bisphosphonates and denosumab in the treatment of pain due to bone metastases: A systematic review within the European Association for Palliative Care guidelines project. *Palliat Med.* 2017 Jan;31(1):5-25. doi: 10.1177/0269216316639793. Epub 2016 Jul 10. PMID: 27006430 (accessed May 2022).

Prostate Cancer UK: Bisphosphonates for advanced prostate cancer. Available online: <https://prostatecanceruk.org/prostate-information/treatments/bisphosphonates> (accessed May 2022).

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Chief Medical Editor, [Professor Tim Iveson](#), Consultant Medical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

Date reviewed

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ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

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